# Syllabus 9758

Statistics Summary

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# 1 Permutations and Combinations

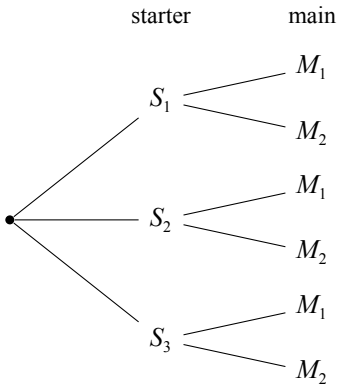
## 1.1 Counting Principles and Formulae

| | Addition principle | Multiplication principle |
|---|---|---|
| Use when | either event $A$ **or** $B$ (cases) | stage 1 **and** stage 2 (steps) |
| Condition | cases cannot happen together | stages are done one after another |
| Total | $m+n$ | $m\times n$ |

| Situation | Formula | Note |
|---|---|---|
| $n$ distinct objects in a row | $n!$ | $0!=1$ |
| $r$ of $n$ distinct objects, in order | ${}^{n}\mathrm{P}_{r}=\dfrac{n!}{(n-r)!}$ | order matters |
| $r$ of $n$ distinct objects, no order | ${}^{n}\mathrm{C}_{r}=\dfrac{n!}{r!\,(n-r)!}$ | ${}^{n}\mathrm{C}_{r}={}^{n}\mathrm{C}_{n-r}$ |
| $r$ places, each of $n$ choices, repeats allowed | $n^{r}$ | objects **can** be reused |
| $n$ objects, $p$ alike of one kind, $q$ alike of another | $\dfrac{n!}{p!\,q!}$ | divide by $k!$ for each $k$ identical |
| $n$ distinct objects in a circle | $(n-1)!$ | rotations are the same |

**GC:** `nPr`, `nCr`: `MATH` → `PRB`. Type $n$ first, then the command, then $r$. `!` is in the same menu.

> **Note**
>
> Order matters (arrange, line up, form a number, assign positions) ⇒ ${}^{n}\mathrm{P}_{r}$. Order does not matter (choose, select, form a committee) ⇒ ${}^{n}\mathrm{C}_{r}$. Selecting then arranging: ${}^{n}\mathrm{C}_{r}\times r!={}^{n}\mathrm{P}_{r}$.



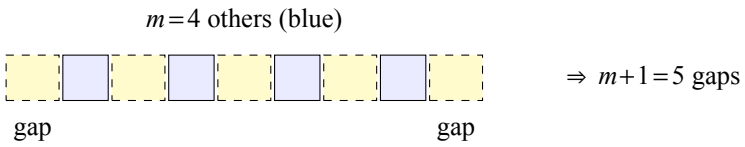

3 starters, 2 mains: each of the 3 starters pairs with each of the 2 mains

$$\text{menus}=3\times 2=6$$

3 starters **or** 2 mains (choose one dish): $3+2=5$

# 1.2 Arrangements in a Line

| Condition | Method | Count |
|---|---|---|
| Fixed position (e.g. $A$ first; number is even; first digit $\neq 0$) | fill the restricted places **first**, then the rest | $\underline{k} \times _ \times \cdots$ |
| $A$, $B$ **together** | glue into one block; arrange the blocks; arrange inside the block | $(n-1)! \times 2!$ |
| $A$, $B$ **not together** | total − together or slot method | $n! - (n-1)!\,2!$ |
| No two of a group adjacent | arrange the others; put the group in the gaps (at most 1 per gap) | $m! \times {}^{m+1}\mathrm{P}_k$ |
| "at least one" | complement: total − none | |
| Several cases (e.g. last digit 0 or not) | case table: count each case, then **add** | |




**Example 1.1**

10 letters of STATISTICS

$$\mathrm{S} \times 3,\ \mathrm{T} \times 3,\ \mathrm{I} \times 2$$

$$\text{total} = \frac{10!}{3!\,3!\,2!}$$
$$= 50\,400$$

**Example 1.2**

4 boys, 3 girls, no two girls adjacent

$$\text{boys} = 4! = 24$$
$$\text{gaps} = 4 + 1 = 5$$
$$\text{girls} = {}^{5}\mathrm{P}_3 = 60$$
$$\therefore \text{total} = 24 \times 60$$
$$= 1440$$

**Example 1.3**

6 people in a row, $A$ and $B$

$$\text{together} = 5! \times 2! = 240$$
$$\text{not together} = 6! - 240$$
$$= 480$$

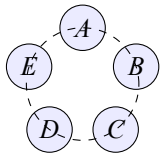
**Example 1.4**

Even 3-digit numbers from $0, 1, \ldots, 5$, no repeats

$$\text{last } 0 = 5 \times 4 = 20$$
$$\text{last } 2 \text{ or } 4 = 2 \times 4 \times 4 = 32$$
$$\therefore \text{total} = 20 + 32$$
$$= 52$$

## 1.3 Circular Arrangements

| Situation | Method | Count |
|---|---|---|
| $n$ people round a table | fix one person; arrange the other $n-1$ | $(n-1)!$ |
| seats **numbered / distinguishable** | seats are different, so rotations differ | $n!$ |
| $A$, $B$ **together** | glue, then circular arrangement of the blocks | $(n-2)!\times 2!$ |
| $A$, $B$ **not together** | total − together | $(n-1)!-(n-2)!\,2!$ |
| $A$, $B$ in fixed relation (e.g. $A$ opposite $B$) | fix $A$; $B$ has 1 place; arrange the rest | $(n-2)!$ |




Rotations are the **same** arrangement.
Clockwise ≠ anticlockwise (different).
5 people: $(5-1)!=24$

**Example 1.5**

6 people round a table, $A$ and $B$ not adjacent

$$\text{adjacent}=(6-2)!\times 2!=48$$

$$\therefore \text{not adjacent}=5!-48=72$$

## 1.4 Selections and Groups

| Situation | Method | Count |
|---|---|---|
| choose $r$ of $n$, with a condition (e.g. ≥ 1 woman) | ${}^{n}\mathrm{C}_{r}$; cases (add), or complement | total − none |
| choose, **then** arrange | select, then multiply by $r!$ (or by the arrangements) | ${}^{n}\mathrm{C}_{r}\times r!$ |
| groups of **different** sizes / labelled groups | choose group by group | ${}^{n}\mathrm{C}_{a}\,{}^{n-a}\mathrm{C}_{b}\cdots$ |
| $k$ groups of the **same** size, unlabelled | as above, then ÷ $k!$ | $\dfrac{{}^{n}\mathrm{C}_{a}\,{}^{n-a}\mathrm{C}_{a}\cdots}{k!}$ |

Are the groups **distinguishable**? Yes (named teams, different tasks, different sizes): no division. No (identical groups of equal size): ÷ $k!$ for $k$ equal groups.

**Example 1.6**

8 people into two groups of 4

$$\text{unlabelled}=\frac{{}^{8}\mathrm{C}_{4}}{2!}=35$$

$$\text{teams } X,\ Y={}^{8}\mathrm{C}_{4}=70$$

**Example 1.7**

Committee of 4 from 6 men, 4 women, at least 1 woman

$$\text{total}={}^{10}\mathrm{C}_{4}=210$$

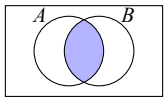
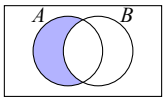
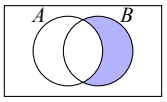
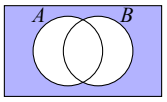
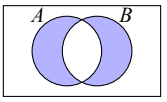
$$\text{no woman}={}^{6}\mathrm{C}_{4}=15$$

$$\therefore \text{ways}=210-15=195$$

# 2 Probability

## 2.1 Basic Rules

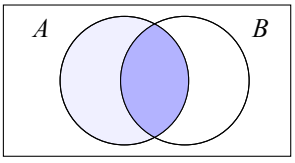
| Result | Formula | Note |
|---|---|---|
| Equally likely outcomes | $\mathrm{P}(A)=\frac{n(A)}{n(S)}$ | $0\leqslant \mathrm{P}(A)\leqslant 1$ |
| Complement | $\mathrm{P}(A')=1-\mathrm{P}(A)$ | "at least one" = 1 – none |
| Addition rule | $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)-\mathrm{P}(A\cap B)$ | $A\cup B$: $A$ or $B$ (or both) |
| Mutually exclusive | $\mathrm{P}(A\cap B)=0$ | so $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$ |
| Exhaustive | $\mathrm{P}(A\cup B)=1$ | |

|  $A\cap B$ |  $A\cap B'$ |  $A'\cap B$ |  $A'\cap B'=(A\cup B)'$ |  exactly one |
|---|---|---|---|---|
| $\mathrm{P}(A\cap B)$ | $\mathrm{P}(A)-\mathrm{P}(A\cap B)$ | $\mathrm{P}(B)-\mathrm{P}(A\cap B)$ | $1-\mathrm{P}(A\cup B)$ | $\mathrm{P}(A)+\mathrm{P}(B)-2\mathrm{P}(A\cap B)$ |

Fill in a Venn diagram from the middle ($A\cap B$) outwards; the four regions add up to 1. $(A\cup B)'=A'\cap B'$ and $(A\cap B)'=A'\cup B'$. None of these probability results is in MF27.

## 2.2 Conditional Probability

| Result | Formula | Note |
|---|---|---|
| Conditional probability | $\mathrm{P}(A\mid B)=\frac{\mathrm{P}(A\cap B)}{\mathrm{P}(B)}$ | probability of $A$ given $B$ has happened; $\mathrm{P}(B)\neq 0$ |
| Multiplication rule | $\mathrm{P}(A\cap B)=\mathrm{P}(A\mid B)\mathrm{P}(B)=\mathrm{P}(B\mid A)\mathrm{P}(A)$ | used along a tree branch |
| Complement given $B$ | $\mathrm{P}(A'\mid B)=1-\mathrm{P}(A\mid B)$ | |
| Warning | $\mathrm{P}(A\mid B)\neq\mathrm{P}(B\mid A)$ | different denominators |

reduced sample space: 

$\mathrm{P}(A\mid B)=\frac{\text{shaded}}{\text{whole circle } B}$

**Example 2.1**

$\mathrm{P}(A)=0.5,\ \mathrm{P}(B)=0.4,\ \mathrm{P}(A\cup B)=0.6$

$$\mathrm{P}(A\cap B)=0.5+0.4-0.6=0.3$$

$$\mathrm{P}(A\mid B)=\frac{0.3}{0.4}=0.75$$

$$\mathrm{P}(A\mid B')=\frac{\mathrm{P}(A)-\mathrm{P}(A\cap B)}{1-\mathrm{P}(B)}=\frac{0.2}{0.6}=\frac{1}{3}$$

## 2.3 Independent and Mutually Exclusive Events

| | **Independent** | **Mutually exclusive** |
|---|---|---|
| Meaning | one event does not change the probability of the other | cannot both happen |
| Test | $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$ | $\mathrm{P}(A\cap B)=0$ |
| Equivalent | $\mathrm{P}(A\mid B)=\mathrm{P}(A)$ and $\mathrm{P}(B\mid A)=\mathrm{P}(B)$ | $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$ |
| Venn | circles overlap, $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$ | circles do not overlap |

**Note**

If $A$, $B$ are independent, so are $A$ and $B'$, $A'$ and $B$, $A'$ and $B'$. If $\mathrm{P}(A)>0$ and $\mathrm{P}(B)>0$, events cannot be both independent and mutually exclusive: $\mathrm{P}(A\cap B)=0\neq\mathrm{P}(A)\mathrm{P}(B)$. To show independence, show $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$; *do not* assume it from the wording.

**Example 2.2**

From Example 2.1: are $A$, $B$ independent?

$$\mathrm{P}(A)\mathrm{P}(B)=0.5\times0.4=0.2$$

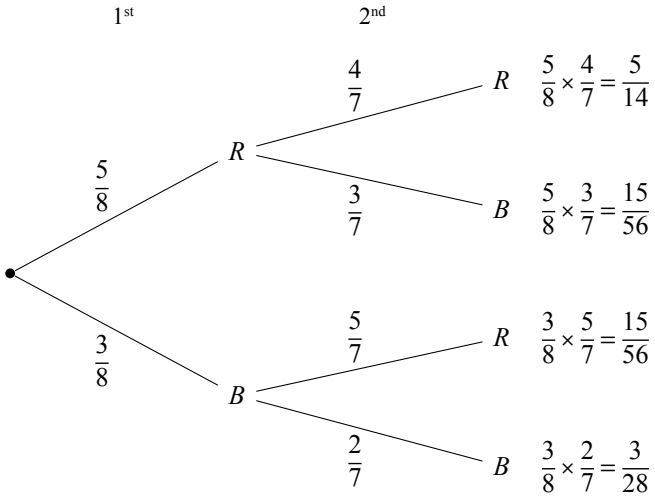
$$\mathrm{P}(A\cap B)=0.3$$

$$\because\ \mathrm{P}(A\cap B)\neq\mathrm{P}(A)\mathrm{P}(B)$$

$$\therefore A,\ B \text{ are not independent}$$

## 2.4 Tree Diagrams

Multiply along branches (∩); add across leaves (∪). Branches from one node sum to 1. Without replacement: probabilities in the second stage **change**.




Bag: 5 red, 3 blue; 2 drawn without replacement.

$$\mathrm{P}(\text{different colours}) = \frac{15}{56} + \frac{15}{56} = \frac{15}{28}$$

$$\mathrm{P}(1\text{st } R \mid 2\text{nd } B) = \frac{\mathrm{P}(R \cap B)}{\mathrm{P}(2\text{nd } B)} = \frac{15/56}{15/56 + 6/56} = \frac{5}{7}$$

Conditional probability from a tree: "given" ⇒ divide by the probability of the *given* event (add its leaves). Bayes' theorem as a named formula is not in the syllabus; use $\mathrm{P}(A \mid B) = \mathrm{P}(A \cap B) / \mathrm{P}(B)$.

## 2.5 Tables of Outcomes and Counting Methods

| + | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 6 | 7 | 8 | 9 | 10 | 11 | 12 |

two dice: $n(S) = 36$

$$\mathrm{P}(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

$$\mathrm{P}(\text{sum} \geqslant 10) = \frac{6}{36} = \frac{1}{6}$$

$$\mathrm{P}(\text{sum} = 7 \mid \text{a 3 shown}) = \frac{2}{11}$$

| Type | Method | Probability |
|---|---|---|
| Selection (no order) | $\frac{\text{ways to choose favourable}}{\text{ways to choose all}}$ using ${}^{n}\mathrm{C}_{r}$ | $\frac{{}^{6}\mathrm{C}_{2}\,{}^{4}\mathrm{C}_{2}}{{}^{10}\mathrm{C}_{4}}$ |
| Arrangement (order) | $\frac{\text{favourable arrangements}}{\text{all arrangements}}$ using $n!$, ${}^{n}\mathrm{P}_{r}$ | e.g. $\frac{2!\,5!}{6!}$ |
| "at least one" | $1-\mathrm{P}(\text{none})$ | |

**Example 2.3**

4 chosen from 6 men, 4 women: $\mathrm{P}(\text{at least 1 woman})$

$$\mathrm{P}(\text{no woman}) = \frac{{}^{6}\mathrm{C}_{4}}{{}^{10}\mathrm{C}_{4}} = \frac{1}{14}$$

$$\therefore \mathrm{P}(\text{at least 1 woman}) = 1 - \frac{1}{14} = \frac{13}{14}$$

## 2.6 Further Question Types

| Type | Method |
|---|---|
| Range of $\mathrm{P}(A \cap B)$ | $\mathrm{P}(A \cap B) \leqslant \min\{\mathrm{P}(A), \mathrm{P}(B)\}$ and<br>$\mathrm{P}(A \cup B) \leqslant 1 \Rightarrow \mathrm{P}(A \cap B) \geqslant \mathrm{P}(A) + \mathrm{P}(B) - 1$ (and $\geqslant 0$) |
| Unknown probability from "independent" / "mutually exclusive" | form an equation: $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$ or $\mathrm{P}(A \cap B) = 0$; solve for the unknown |
| Games: "first to win", taking turns | list the winning sequences; the probabilities form a **GP**; $S_{\infty} = \frac{a}{1-r}$ $(\lvert r \rvert < 1)$ |
| Proving a probability statement | sketch the Venn diagram; label regions with $\mathrm{P}(\cdot)$; use $\mathrm{P}(\cdot) \geqslant 0$ for every region |

**Example 2.4**

$\mathrm{P}(A) = 0.7,\ \mathrm{P}(B) = 0.6$

$$\mathrm{P}(A \cap B) \leqslant 0.6$$

$$\mathrm{P}(A \cap B) \geqslant 0.7 + 0.6 - 1 = 0.3$$

$$\therefore 0.3 \leqslant \mathrm{P}(A \cap B) \leqslant 0.6$$

**Example 2.5**

$A$, $B$ take turns to roll a die, $A$ first; first six wins

$$\mathrm{P}(A \text{ wins}) = \frac{1}{6} + \left(\frac{5}{6}\right)^{2}\frac{1}{6} + \left(\frac{5}{6}\right)^{4}\frac{1}{6} + \cdots$$

$$= \frac{1/6}{1 - 25/36}$$

$$= \frac{6}{11}$$

# 3 Discrete Random Variables

## 3.1 Probability Distribution

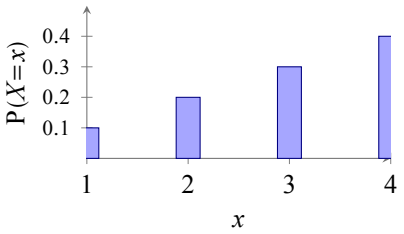
| Item | Result | Note |
|---|---|---|
| Discrete r.v. $X$ | takes separate values $x_1, x_2, \ldots$ (counts) | e.g. number of heads |
| Probability distribution | table of $x$ against $\mathrm{P}(X=x)$ | |
| Conditions | $0 \leqslant \mathrm{P}(X=x) \leqslant 1$ and $\sum_{\text{all } x} \mathrm{P}(X=x) = 1$ | use to find an unknown constant |
| $\mathrm{P}(X \leqslant a)$ | add the probabilities of all $x \leqslant a$ | no cumulative function needed |

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $\mathrm{P}(X=x)$ | $k$ | $2k$ | $3k$ | $4k$ |

$$\sum \mathrm{P}(X=x) = 1$$

$$10k = 1$$

$$\therefore k = 0.1$$




## 3.2 Expectation and Variance

| Result | Formula | Note |
|---|---|---|
| Expectation (mean) | $\mathrm{E}(X) = \mu = \sum x\,\mathrm{P}(X=x)$ | long-run average |
| Expectation of a function | $\mathrm{E}(g(X)) = \sum g(x)\,\mathrm{P}(X=x)$ | e.g. $\mathrm{E}(X^2) = \sum x^2 \mathrm{P}(X=x)$ |
| Variance | $\mathrm{Var}(X) = \mathrm{E}[(X-\mu)^2] = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$ | $\mathrm{Var}(X) \geqslant 0$ |
| Standard deviation | $\sigma = \sqrt{\mathrm{Var}(X)}$ | |

**GC:** enter $x$ in **L1**, $\mathrm{P}(X=x)$ in **L2**. **STAT** → **CALC** → **1-Var Stats L1,L2**. Then $\bar{x} = \mathrm{E}(X)$, $\sigma\mathbf{x} = \sigma$ (not **Sx**); $\mathrm{Var}(X) = \sigma^2$. $\mathrm{E}(X^2) = \mathrm{Var}(X) + [\mathrm{E}(X)]^2$, or read $\Sigma\mathbf{x}^2$.

$$\mathrm{E}(X) = 1(0.1) + 2(0.2) + 3(0.3) + 4(0.4) = 3$$

$$\mathrm{E}(X^2) = 1(0.1) + 4(0.2) + 9(0.3) + 16(0.4) = 10$$

$$\mathrm{Var}(X) = 10 - 3^2 = 1$$

## 3.3 Linear Functions and Combinations

| Result | Formula | Condition |
|---|---|---|
| Linear function | $\mathrm{E}(aX+b)=a\mathrm{E}(X)+b$ | always |
| | $\mathrm{Var}(aX+b)=a^2\mathrm{Var}(X)$ | always; $b$ does not change spread |
| Sum / difference | $\mathrm{E}(aX\pm bY)=a\mathrm{E}(X)\pm b\mathrm{E}(Y)$ | always |
| | $\mathrm{Var}(aX\pm bY)=a^2\mathrm{Var}(X)+b^2\mathrm{Var}(Y)$ | $X$, $Y$ **independent**; variances **add** for $\pm$ |

| | $X_1+X_2+\cdots+X_n$ | $nX$ |
|---|---|---|
| Meaning | $n$ **independent** observations of $X$ | one observation, multiplied by $n$ |
| Mean | $n\mu$ | $n\mu$ |
| Variance | $n\sigma^2$ | $n^2\sigma^2$ |

**Example 3.1**

$Y=3X-2$, using $\mathrm{E}(X)=3$, $\mathrm{Var}(X)=1$

$$\mathrm{E}(Y)=3(3)-2=7$$

$$\mathrm{Var}(Y)=3^2(1)=9$$

**Example 3.2**

Prize $W$: $0, 4, 10$ with probabilities $0.6, 0.3, 0.1$; cost to play \$3

$$\mathrm{E}(W)=0+1.2+1=2.2$$

$$\mathrm{E}(\text{gain})=2.2-3=-0.8$$

$\therefore$ not a fair game

Fair game: $\mathrm{E}(\text{gain})=0$.

# 4 Binomial Distribution

## 4.1 The Binomial Model

| Item | Result | Note |
|---|---|---|
| Notation | $X \sim \mathrm{B}(n,p)$ | $X$ = no. of successes in $n$ trials |
| Probability | $\mathrm{P}(X=r) = {}^{n}\mathrm{C}_{r}\, p^{r}(1-p)^{n-r}, \quad r=0,1,\ldots,n$ | MF27 |
| Mean | $\mathrm{E}(X) = np$ | MF27 (no proof) |
| Variance | $\mathrm{Var}(X) = np(1-p)$ | MF27 (no proof) |

**Conditions for $X \sim \mathrm{B}(n,p)$ (state in context)**

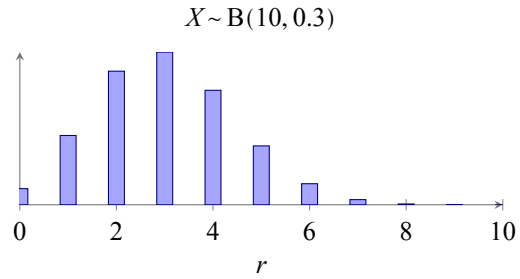
- fixed number $n$ of trials
- each trial has exactly 2 outcomes: success / failure
- $\mathrm{P}(\text{success}) = p$ is the **same** for every trial
- trials are **independent**

| Model fails when … | Why |
|---|---|
| selecting without replacement from a small population | $p$ changes; trials not independent |
| counting trials until the first success | $n$ is not fixed |
| events are linked (e.g. family members, same day) | trials not independent |
| $p$ varies (e.g. by time of day, by type of item) | $p$ not constant |

## 4.2 Using the GC

| Probability | GC command | Note |
|---|---|---|
| $\mathrm{P}(X=r)$ | `binompdf(n,p,r)` | |
| $\mathrm{P}(X \leqslant r)$ | `binomcdf(n,p,r)` | adds from 0 to $r$ |
| $\mathrm{P}(X<r)$ | `binomcdf(n,p,r-1)` | |
| $\mathrm{P}(X \geqslant r)$ | 1–`binomcdf(n,p,r-1)` | |
| $\mathrm{P}(X>r)$ | 1–`binomcdf(n,p,r)` | |
| $\mathrm{P}(a \leqslant X \leqslant b)$ | `binomcdf(n,p,b)`–`binomcdf(n,p,a-1)` | the lower bound is $a-1$ |

**GC:** 2nd VARS (DISTR) → `binompdf(` / `binomcdf(`.




**Example 4.1**

$X \sim \mathrm{B}(10, 0.3)$

$$\mathrm{P}(X=3) = {}^{10}\mathrm{C}_3 (0.3)^3 (0.7)^7$$
$$= 0.267$$
$$\mathrm{P}(X \geqslant 4) = 1 - \mathrm{P}(X \leqslant 3)$$
$$= 1 - 0.6496 = 0.350$$
$$\mathrm{E}(X) = 3, \ \mathrm{Var}(X) = 2.1$$

## 4.3 Finding $n$ or $p$; Combining with Other Results

| Type | Method |
|---|---|
| Smallest $n$ with $\mathrm{P}(X \geqslant 1) > c$ | $1-(1-p)^n > c \Rightarrow (1-p)^n < 1-c$; take ln (dividing by $\ln(1-p) < 0$ **reverses** the inequality) |
| Other inequalities in $n$ | GC table: $\mathrm{Y}_1 =$ `binomcdf(n,p,r)` with `X` as $n$; read the first $n$ that satisfies the condition |
| Unknown $p$ | form the equation in $p$ (e.g. $\mathrm{P}(X=0) = (1-p)^n$); solve; or GC graph/table in $p$ |
| Given a condition | conditional probability: $\mathrm{P}(X=r \mid X \geqslant 1) = \dfrac{\mathrm{P}(X=r)}{\mathrm{P}(X \geqslant 1)}$, $r \geqslant 1$ |
| Binomial within binomial | new $p'$ from the first part, e.g. $Y \sim \mathrm{B}(m, p')$ where $p' = \mathrm{P}(X \geqslant 2)$ ("on how many of $m$ days") |

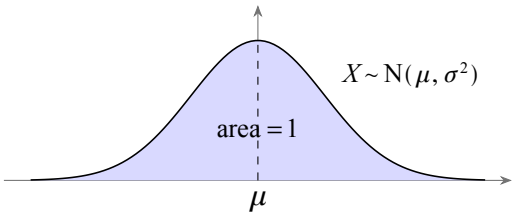
**Example 4.2**

$\mathrm{P}(X \geqslant 1) > 0.99$ for $X \sim \mathrm{B}(n, 0.1)$

$$1 - 0.9^n > 0.99$$
$$0.9^n < 0.01$$
$$n \ln 0.9 < \ln 0.01$$
$$n > \frac{\ln 0.01}{\ln 0.9} = 43.7$$
$$\therefore n_{\min} = 44$$

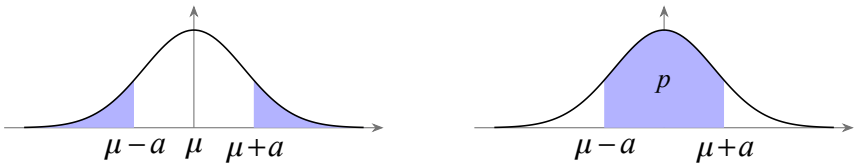
# 5 Normal Distribution

## 5.1 Properties




- bell-shaped, symmetric about $x = \mu$
- total area = 1; mean = $\mu$; variance = $\sigma^2$
- continuous: $\mathrm{P}(X = a) = 0$, so $\mathrm{P}(X < a) = \mathrm{P}(X \leqslant a)$
- $\sigma$ larger ⇒ wider and flatter

| Item | Result | Note |
|---|---|---|
| Standardise | $Z = \dfrac{X-\mu}{\sigma} \sim \mathrm{N}(0, 1)$ | $Z$ = number of s.d. from the mean |
| Symmetry | $\mathrm{P}(X < \mu - a) = \mathrm{P}(X > \mu + a)$ | $\mathrm{P}(Z < -z) = \mathrm{P}(Z > z) = 1 - \mathrm{P}(Z < z)$ |
| Complement | $\mathrm{P}(X > a) = 1 - \mathrm{P}(X < a)$ | total area is 1 |




Left: equal tails. Right: $\mathrm{P}(\mu - a < X < \mu + a) = p$, so each tail is $\frac{1-p}{2}$.

## 5.2 Finding Probabilities and Values (GC)

| Find | GC command | Note |
|---|---|---|
| $\mathrm{P}(a < X < b)$ | `normalcdf(a,b,`$\mu$`,`$\sigma$`)` | enter $\sigma$, **not** $\sigma^2$ |
| $\mathrm{P}(X < a)$ | `normalcdf(-1E99,a,`$\mu$`,`$\sigma$`)` | 1 2nd , 9 9 for 1E99 |
| $\mathrm{P}(X > a)$ | `normalcdf(a,1E99,`$\mu$`,`$\sigma$`)` | |
| $x$ with $\mathrm{P}(X < x) = p$ | `invNorm(p,`$\mu$`,`$\sigma$`)` | `invNorm` gives the **left** area |
| $x$ with $\mathrm{P}(X > x) = p$ | `invNorm(1-p,`$\mu$`,`$\sigma$`)` | |
| $x$ with $\mathrm{P}(\mu - c < X < \mu + c) = p$ | $x = \mu + c =$ `invNorm(`$\frac{1+p}{2}$`,`$\mu$`,`$\sigma$`)` | each tail is $\frac{1-p}{2}$ |

**GC:** 2nd VARS (DISTR) → `normalcdf(`, `invNorm(`. If $\mu, \sigma$ are omitted, the GC uses $\mathrm{N}(0, 1)$.

**Example 5.1**

$X \sim \mathrm{N}(60, 8^2)$: find $\mathrm{P}(52 < X < 70)$, and $x$ with $\mathrm{P}(X > x) = 0.2$

$$\mathrm{P}(52 < X < 70) = 0.736$$

$$\mathrm{P}(X < x) = 0.8$$

$$x = \mathrm{invNorm}(0.8, 60, 8)$$

$$= 66.7$$

**Example 5.2**

Find $\mu$, $\sigma$ given $\mathrm{P}(X < 30) = 0.1$ and $\mathrm{P}(X > 46) = 0.2$

$$\frac{30 - \mu}{\sigma} = \mathrm{invNorm}(0.1) = -1.2816$$

$$\frac{46 - \mu}{\sigma} = \mathrm{invNorm}(0.8) = 0.8416$$

$$\frac{16}{\sigma} = 2.1232$$

$$\sigma = 7.54$$

$$\mu = 39.7$$

## 5.3 Linear Functions and Combinations of Normals

| Situation | Distribution | Condition |
|---|---|---|
| $aX+b$ | $\mathrm{N}(a\mu+b, a^2\sigma^2)$ | $X$ normal |
| $aX \pm bY$ | $\mathrm{N}(a\mu_X \pm b\mu_Y, a^2\sigma_X^2+b^2\sigma_Y^2)$ | $X$, $Y$ independent normals |
| $X_1+X_2+\cdots+X_n$ | $\mathrm{N}(n\mu, n\sigma^2)$ | independent copies of $X$ |
| $nX$ | $\mathrm{N}(n\mu, n^2\sigma^2)$ | one observation $\times n$ |

Common mistake

× Variances **add** for both $X+Y$ and $X-Y$: $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$. $\mathrm{Var}(2X) = 4\mathrm{Var}(X) \neq \mathrm{Var}(X_1+X_2) = 2\mathrm{Var}(X)$. A comparison such as $X > Y$ is rewritten as $X - Y > 0$, then one normal variable.

**Example 5.3**

$X \sim \mathrm{N}(120, 15^2)$, $Y \sim \mathrm{N}(100, 8^2)$ independent; $\mathrm{P}(X > Y)$

$$X - Y \sim \mathrm{N}(20, 15^2+8^2)$$

$$= \mathrm{N}(20, 17^2)$$

$$\mathrm{P}(X > Y) = \mathrm{P}(X - Y > 0)$$

$$= 0.880$$

A normal model may be unsuitable if the quantity cannot be negative but $\mu$ is small compared with $\sigma$ (gives $\mathrm{P}(X < 0) > 0$), if the data are skewed or bounded, or if observations are not independent. Normal approximation to a binomial distribution is not in the H2 syllabus.

# 6 Sampling and the Central Limit Theorem

## 6.1 Population and Sample

| Term | Meaning | Notation |
|---|---|---|
| Population | the whole group of interest | mean $\mu$, variance $\sigma^2$ (parameters) |
| Sample | a part of the population, size $n$ | $\bar{x}$, $s^2$ (statistics) |
| Simple random sample | every member (and every sample of size $n$) is equally likely to be chosen; members chosen independently | |
| To obtain one | list and number the whole population; choose numbers by a random number generator; ignore repeats | |

| Not random if … | Reason |
|---|---|
| chosen by convenience (e.g. first 30 in the canteen, volunteers) | some members are more likely than others |
| only part of the population can be chosen | the rest has no chance |
| one choice affects another (e.g. friends chosen together) | not independent |

## 6.2 Unbiased Estimates

| Estimate of | Raw data | Coded: $\Sigma(x-a)$, $\Sigma(x-a)^2$ |
|---|---|---|
| Mean $\mu$ | $\bar{x}=\dfrac{\sum x}{n}$ | $\bar{x}=a+\dfrac{\sum(x-a)}{n}$ |
| Variance $\sigma^2$ | $s^2=\dfrac{1}{n-1}\left[\sum x^2-\dfrac{\left(\sum x\right)^2}{n}\right]$ | $s^2=\dfrac{1}{n-1}\left[\sum(x-a)^2-\dfrac{\left(\sum(x-a)\right)^2}{n}\right]$ |
| Also | $s^2=\dfrac{1}{n-1}\sum(x-\bar{x})^2$ | MF27 |

**GC:** STAT → CALC → 1-Var Stats. Use Sx (=$s$, divides by $n-1$), so $s^2=$Sx$^2$. Do **not** use $\sigma$x (divides by $n$; this is not the unbiased estimate).

**Example 6.1**

$n=10, \sum x=125, \sum x^2=1830$

$$\bar{x}=\frac{125}{10}=12.5$$

$$s^2=\frac{1}{9}\left[1830-\frac{125^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

**Example 6.2**

given $\sum(x-10)=25, \sum(x-10)^2=330$

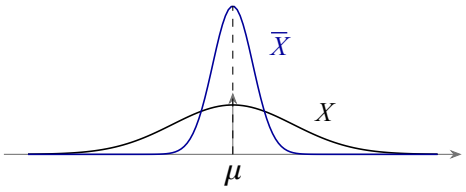
$$\bar{x}=10+\frac{25}{10}=12.5$$

$$s^2=\frac{1}{9}\left[330-\frac{25^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

## 6.3 Distribution of the Sample Mean

| Population | Distribution of $\overline{X}$ | Reason |
|---|---|---|
| Normal, $X\sim \mathrm{N}(\mu,\sigma^2)$ | $\overline{X}\sim \mathrm{N}(\mu,\frac{\sigma^2}{n})$ (exact) | any $n$; linear combination of normals |
| Not normal / unknown, $n$ **large** (e.g. $n\geqslant 30$) | $\overline{X}\approx \mathrm{N}(\mu,\frac{\sigma^2}{n})$ (approx.) | **Central Limit Theorem** |
| Any population | $\mathrm{E}(\overline{X})=\mu,\ \mathrm{Var}(\overline{X})=\frac{\sigma^2}{n}$ | true for every $n$ |




Same centre $\mu$. $\overline{X}$ is **narrower**: s.d. $\frac{\sigma}{\sqrt{n}}$ (larger $n$, narrower).

**Example 6.3**

population mean 80, s.d. 12, $n=40$

$$\overline{X}\approx \mathrm{N}(80,\frac{12^2}{40}) \text{ (CLT)}$$

$$\mathrm{P}(\overline{X}>83)=0.0569$$

**Central Limit Theorem**

For a random sample of size $n$ (large, e.g. $n\geqslant 30$) from **any** population with mean $\mu$ and variance $\sigma^2$: $\overline{X}\approx \mathrm{N}(\mu,\frac{\sigma^2}{n})$

Likewise $\sum_{i=1}^{n} X_i \approx \mathrm{N}(n\mu, n\sigma^2)$. If $n$ is small, the population must be normal.

## 6.4 Which Distribution?

| The question … | Use | Key step |
|---|---|---|
| number of successes in $n$ independent trials | $\mathrm{B}(n,p)$ | state the conditions |
| one measurement, normal | $\mathrm{N}(\mu,\sigma^2)$ | standardise / GC |
| total or difference of independent normals | $\mathrm{N}(\cdot,\cdot)$ | add the **variances** |
| mean of $n$ values, population normal | $\overline{X}\sim \mathrm{N}(\mu,\sigma^2/n)$ | exact |
| mean of $n$ values, population not normal, $n$ large | $\overline{X}\approx \mathrm{N}(\mu,\sigma^2/n)$ | say "by CLT" |

# 7 Hypothesis Testing

## 7.1 Terminology

| Term | Meaning |
|---|---|
| Null hypothesis $H_0$ | the claim assumed true: $H_0: \mu = \mu_0$ |
| Alternative hypothesis $H_1$ | what we look for evidence of: $\mu < \mu_0$, $\mu > \mu_0$ (1-tail) or $\mu \neq \mu_0$ (2-tail) |
| Test statistic | $\bar{X}$ (or $Z$), whose distribution is found assuming $H_0$ is true |
| Level of significance $\alpha$ | probability of rejecting $H_0$ when $H_0$ is true; given in the question |
| Critical region | values of the test statistic for which $H_0$ is rejected; boundary = critical value |
| $p$-value | probability, assuming $H_0$ true, of a test statistic at least as extreme as the one observed; smallest level at which $H_0$ is rejected |

## 7.2 Formulating the Hypotheses

| Wording | $H_1$ | Type |
|---|---|---|
| decreased, less than, lower, overstated | $\mu < \mu_0$ | 1-tail (lower) |
| increased, more than, greater, understated | $\mu > \mu_0$ | 1-tail (upper) |
| changed, different, not correct, biased | $\mu \neq \mu_0$ | 2-tail |

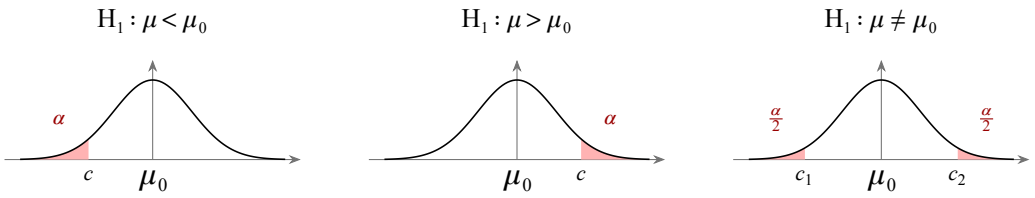
Define $\mu$ in context (e.g. "$\mu$ is the population mean mass"). $H_0$ always uses "=".

## 7.3 Test Statistic and Its Distribution

| Situation | Distribution under $H_0$ | Say |
|---|---|---|
| Normal population, $\sigma^2$ known | $\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$ | any $n$ |
| Any population, $n$ large, $\sigma^2$ known | $\bar{X} \approx N(\mu_0, \frac{\sigma^2}{n})$ | "by CLT" |
| Any population, $n$ large, $\sigma^2$ unknown | $\bar{X} \approx N(\mu_0, \frac{s^2}{n})$ | "by CLT"; $s^2$ unbiased estimate |

Random sample assumed. Small $n$ with unknown $\sigma^2$ is not tested in H2 (no $t$-test). Small $n$ needs "population is normal" to be given or assumed.

## 7.4 The Test: *p*-value and Critical Region




| | **Lower tail** | **Upper tail** | **2-tail** |
|---|---|---|---|
| *p*-value | $P(\bar{X} \leqslant \bar{x})$ | $P(\bar{X} \geqslant \bar{x})$ | $2P(\bar{X} \leqslant \bar{x})$ if $\bar{x} < \mu_0$;<br>$2P(\bar{X} \geqslant \bar{x})$ if $\bar{x} > \mu_0$ |
| Reject $H_0$ if | *p*-value $\leqslant \alpha$ | | |
| Critical region | $\bar{X} \leqslant c$, $P(\bar{X} \leqslant c) = \alpha$ | $\bar{X} \geqslant c$, $P(\bar{X} \geqslant c) = \alpha$ | $\bar{X} \leqslant c_1$ or $\bar{X} \geqslant c_2$; each tail $\frac{\alpha}{2}$ |
| Reject $H_0$ if | $\bar{x}$ lies in the critical region | | |

| **Critical *z*** | $\alpha = 5\%$ | $\alpha = 1\%$ | $\alpha = 10\%$ | $\alpha = 2\%$ |
|---|---|---|---|---|
| 1-tail (upper) | 1.645 | 2.326 | 1.282 | 2.054 |
| 2-tail | $\pm 1.960$ | $\pm 2.576$ | $\pm 1.645$ | $\pm 2.326$ |

Using $z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$: reject $H_0$ if $z$ is in the critical region (lower tail: $z \leqslant -1.645$). The same decision as the *p*-value test.

**Steps of a test**

1 $H_0: \mu = \mu_0$, $H_1: \mu \square \mu_0$; define $\mu$; state $\alpha$
2 distribution of $\bar{X}$ under $H_0$ (with reason: normal population / CLT)
3 *p*-value from the GC (or critical region)
4 compare: $p \leqslant \alpha \Rightarrow$ reject $H_0$; $p > \alpha \Rightarrow$ do not reject $H_0$
5 conclusion **in context**

| **Result** | **Conclusion (in context)** |
|---|---|
| Reject $H_0$ | there is sufficient evidence, at the $\alpha$ level of significance, to conclude that [$H_1$ in words] |
| Do not reject $H_0$ | there is insufficient evidence, at the $\alpha$ level of significance, to conclude that [$H_1$ in words] |

Never write "accept $H_0$" or "$H_0$ is true / proven".

**GC:** STAT → TESTS → 1:Z-Test. Inpt:Stats; enter $\mu 0$, $\sigma$ (use $s$ if $\sigma$ unknown, large $n$), $\bar{x}$, n; choose $\neq \mu 0$, $< \mu 0$ or $> \mu 0$ to match $H_1$; Calculate gives $z$ and $p$. With raw data use Inpt:Data with the list.

**Example 7.1**

$H_0: \mu = 500$, $H_1: \mu < 500$; $\sigma = 10$, $n = 50$, $\bar{x} = 497$, $\alpha = 5\%$

$$\overline{X} \sim N(500, \frac{10^2}{50}) \text{ under } H_0 \text{ (CLT)}$$

$$p\text{-value} = P(\overline{X} \leqslant 497) = 0.0169$$

$$\because 0.0169 \leqslant 0.05$$

$$\therefore \text{reject } H_0$$

**Example 7.2**

$H_0: \mu = 25$, $H_1: \mu \neq 25$; $n = 36$, $\bar{x} = 26.2$, $s^2 = 9$

$$\overline{X} \approx N(25, \frac{9}{36}) \text{ under } H_0 \text{ (CLT)}$$

$$z = \frac{26.2 - 25}{0.5} = 2.4$$

$$p\text{-value} = 2P(\overline{X} \geqslant 26.2) = 0.0164$$

$$0.0164 > 0.01 \Rightarrow \text{do not reject } H_0 \quad (\alpha = 1\%)$$

$$0.0164 \leqslant 0.05 \Rightarrow \text{reject } H_0 \quad (\alpha = 5\%)$$

## 7.5 Using the Result of a Test to Find Unknowns

| Given | Method |
|---|---|
| $H_0$ rejected at level $\alpha$; find range of $\alpha$ | $\alpha \geqslant p$-value (smallest level of significance $= p$-value) |
| Range of $\bar{x}$ (or $\mu_0$, $n$) for which $H_0$ is rejected | critical region: find $c$ by **invNorm**, then write the inequality in $\bar{x}$ |
| Range of $n$ | $p$-value (or $z$) in terms of $n$; use a GC table in $n$ to find where the condition holds |

**Example 7.3**

$H_0: \mu = 50$, $H_1: \mu > 50$; $\sigma = 8$, $n = 40$, $\alpha = 5\%$. Critical region?

$$\overline{X} \sim N(50, \frac{8^2}{40})$$

$$P(\overline{X} \geqslant c) = 0.05$$

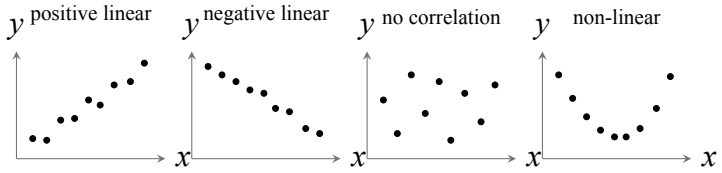
$$c = \text{invNorm}(0.95, 50, \sqrt{1.6})$$

$$= 52.08$$

$$\therefore \text{reject } H_0 \text{ if } \bar{x} \geqslant 52.08$$

# 8 Correlation and Linear Regression

## 8.1 Scatter Diagrams




| Item | Note |
|---|---|
| Independent $x$, dependent $y$ | $x$: controlled / explains; $y$: the response (e.g. $y$ = sales, $x$ = advertising) |
| Describe a scatter diagram | direction (positive / negative), linear or curved, strength (close to a line), outliers |
| Outlier | point far from the pattern; check data, comment on its effect on $r$ and on the line |
| GC | enter $x$ in **L1**, $y$ in **L2**; 2nd Y= (STAT PLOT) → **Plot1** on, scatter; ZOOM **9:ZoomStat** |

## 8.2 Product Moment Correlation Coefficient $r$

**PMCC (MF27)**

$$r=\frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}=\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

$$S_{xx}=\sum x^2-\frac{\left(\sum x\right)^2}{n},\quad S_{yy}=\sum y^2-\frac{\left(\sum y\right)^2}{n},\quad S_{xy}=\sum xy-\frac{\sum x\sum y}{n}$$

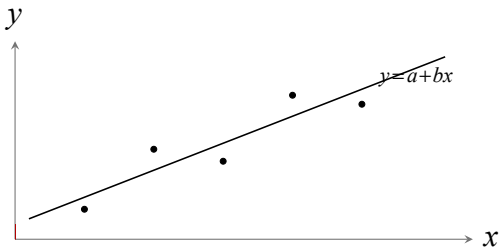
| Value of $r$ | Interpretation |
|---|---|
| $r=1$ or $-1$ | all points lie exactly on a straight line (positive / negative gradient) |
| $r$ close to 1 | strong positive linear correlation |
| $r$ close to $-1$ | strong negative linear correlation |
| $r$ close to 0 | little or no **linear** correlation (a non-linear relation may still exist) |

Note

$-1\leqslant r\leqslant 1$. $|r|$ close to 1 only says a **linear** model fits well; check the scatter diagram. A strong correlation does **not** mean that $x$ causes $y$.

**GC:** 2nd 0 (CATALOG) → **DiagnosticOn** (once). Then STAT → **CALC** → **4:LinReg(ax+b) L1,L2** gives **a**, **b**, **r**. Use **8:LinReg(a+bx)** for $y=a+bx$.

## 8.3 Least-Squares Regression Line




Line of $y$ on $x$ minimises $\sum$(vertical distances)$^2$ (red).

| | $y$ **on** $x$ | $x$ **on** $y$ |
|---|---|---|
| Equation | $y-\bar{y}=b(x-\bar{x}),\ b=\frac{S_{xy}}{S_{xx}}$ | $x-\bar{x}=d(y-\bar{y}),\ d=\frac{S_{xy}}{S_{yy}}$ |
| Or | $y=a+bx,\ a=\bar{y}-b\bar{x}$ | $x=c+dy,\ c=\bar{x}-d\bar{y}$ |
| Use to estimate | $y$ from a given $x$ | $x$ from a given $y$ |

**Note**

Both lines pass through $(\bar{x},\bar{y})$. To estimate $x$ from $y$, use the line of $x$ on $y$ (do **not** rearrange $y$ on $x$), whichever variable is the independent one. The gradient $b$ has the sign of $r$.

**Example 8.1**

$x$: 1, 2, 3, 4, 5, 6 $y$: 3.1, 4.3, 5.2, 6.8, 7.1, 8.9

$$\sum x=21,\ \sum y=35.4,\ \sum xy=143.4,\ \sum x^2=91,\ \sum y^2=231$$

$$S_{xx}=91-\frac{21^2}{6}=17.5$$

$$S_{xy}=143.4-\frac{21(35.4)}{6}=19.5$$

$$S_{yy}=231-\frac{35.4^2}{6}=22.14$$

$$r=\frac{19.5}{\sqrt{17.5\times 22.14}}=0.991$$

$$b=\frac{19.5}{17.5}=1.114,\quad a=5.9-1.114(3.5)=2.00$$

$$\therefore y=2.00+1.11x$$

## 8.4 Interpolation, Extrapolation and Interpretation

| Item | Note |
|---|---|
| Interpolation | estimate for $x$ **within** the data range: reliable if $\|r\|$ is close to 1 and the scatter diagram is linear |
| Extrapolation | estimate for $x$ **outside** the data range: unreliable, the linear relation may not continue |
| Gradient $b$ | the change in $y$ per unit increase in $x$ (in context) |
| Intercept $a$ | value of $y$ when $x=0$; meaningful only if $x=0$ is sensible in context |
| How well does the model fit? | $\|r\|$ close to 1 and points close to the line ⇒ good fit; comment on the range of $x$ |

**Example 8.2**

using the example above: $x=3.5$ and $x=10$

$$x=3.5: \ y=2.00+1.114(3.5)=5.90 \ \text{(interpolation, reliable)}$$

$$x=10: \ y=2.00+1.114(10)=13.1 \ \text{(extrapolation, unreliable)}$$

## 8.5 Transformations to Achieve Linearity

Choose the model whose scatter diagram of (transformed variable, $y$) is the most linear, and whose $|r|$ is closest to 1.

| Model | Scatter diagram | Plot | Regression | Shape |
|---|---|---|---|---|
| $y=a+bx^2$ |  | $y$ against $x^2$ | $y$ on $x^2$ | increasing, curving up |
| $y=a+b\ln x$ |  | $y$ against $\ln x$ | $y$ on $\ln x$ | increasing, curving down |
| $y=a+\frac{b}{x}$ |  | $y$ against $\frac{1}{x}$ | $y$ on $\frac{1}{x}$ | decreasing, levelling off |
| $y=a\,e^{bx}$ |  | $\ln y$ against $x$ | $\ln y$ on $x$ | increasing, curving up faster |

Let $w$ be the transformed variable (e.g. $w=x^2$, $\ln x$, $\frac{1}{x}$). Find $r$ and the line of $y$ on $w$ with the GC (new list), then substitute $w=\ldots$ to estimate $y$. For $\ln y$ on $x$: $\ln y=a+bx \Rightarrow y=e^a e^{bx}$.